

Project Analysis: Fraud Detection in Bank Transactions

1. Main Objective

Build a system that:

- ☐ **Analyzes bank transactions**
- ☐ **detects suspicious behavior,**
- ☐ **calculates a fraud risk score,**
- ☐ **explains the reason,**
- ☐ **recommends an action.**

2. Core Project Features

A. Transaction Analysis

Analyze:

- Transaction amount
- Transaction time
- Transaction type
- Merchant
- Location
- Device
- IP address
- Payment method

B. Customer Behavioral Analysis

Compare the current transaction with historical behavior:

- Average transaction amount
- Maximum transaction amount
- Number of transactions
- Transaction frequency
- Usual transaction time
- Usual location
- Usual device

3.Feature Engineering

- ☐ amount_deviation
- ☐ transactions_last_10_minutes
- ☐ transactions_last_1_hour
- ☐ transactions_last_24_hours
- ☐ average_customer_amount
- ☐ is_new_device
- ☐ is_new_location
- ☐ time_since_last_transaction
- ☐ distance_from_previous_location
- ☐ is_unusual_time

4. Detection Methods

1. Rule-Based Detection

Example:

- ☐ If amount $> 10 \times$ average amount
→ Increase risk
- ☐ If new device
→ Increase risk
- ☐ If many transactions in a short time
→ Increase risk

Advantages:

- Simple
- Fast
- Easy to explain

2. Supervised Machine Learning

Train using labeled data:

0 → Legitimate

1 → Fraud

3. Anomaly Detection

Detects unusual behavior that was not present in the training data.

Possible algorithms:

- Isolation Forest
- Autoencoder

Models to compare:

- Logistic Regression
- Random Forest
- XGBoost / LightGBM

5. Hybrid System

Rule Score+ML Fraud Probability+Anomaly Score

Calculate: **Final Risk Score = Combined Score**

Ex.

0–30 → Low Risk → Approve

31–70 → Medium Risk → Monitor / Verify

71–100 → High Risk → Review / Block

6. Explainable AI

The system should explain: **Why was this transaction flagged?**

Ex.:

Risk Score: 91/100

Reasons:

- Transaction amount is 8× higher than normal
- New device detected
- New location detected
- Multiple transactions within 5 minutes

Tool: [SHAP](#)

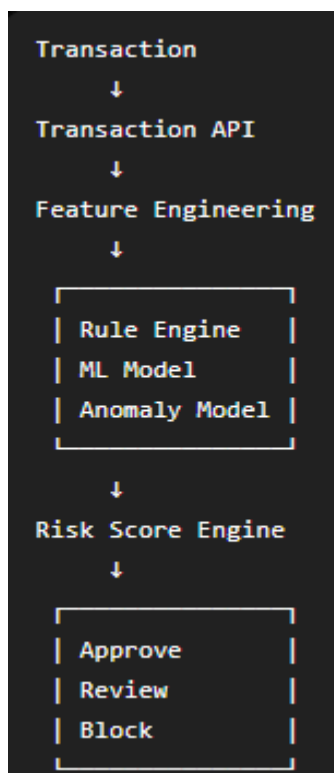
8. Fraud Detection Metrics

- Precision
- Recall
- F1-Score
- PR-AUC
- ROC-AUC
- Confusion Matrix
- False Positive Rate
- False Negative Rate

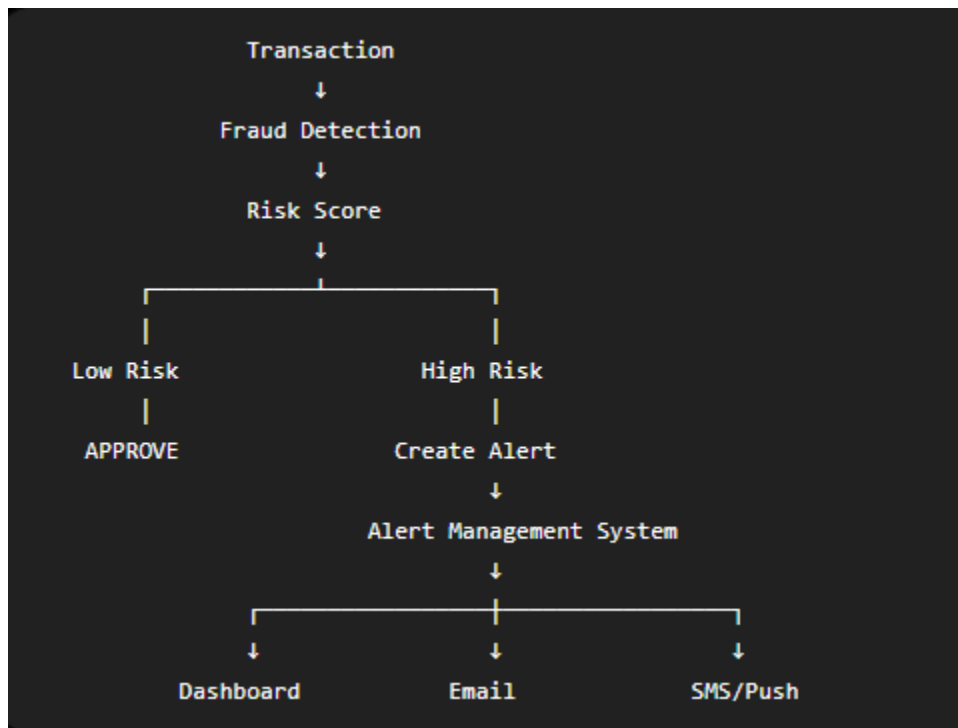
8. Financial Cost Analysis

- False Negative= Actual fraud not detected= Financial loss
- False Positive= Legitimate transaction incorrectly flagged= Customer inconvenience / investigation cost

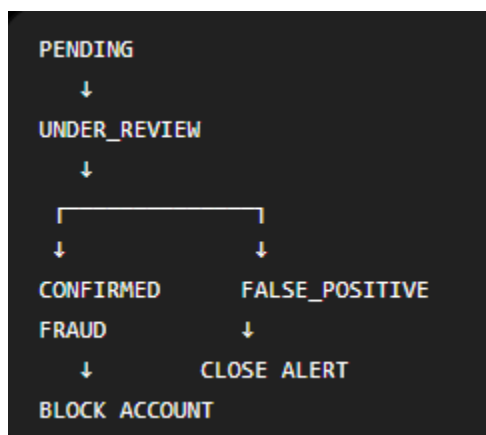
9. Real-Time System Architecture



10.Alert Architecture



Alert Status



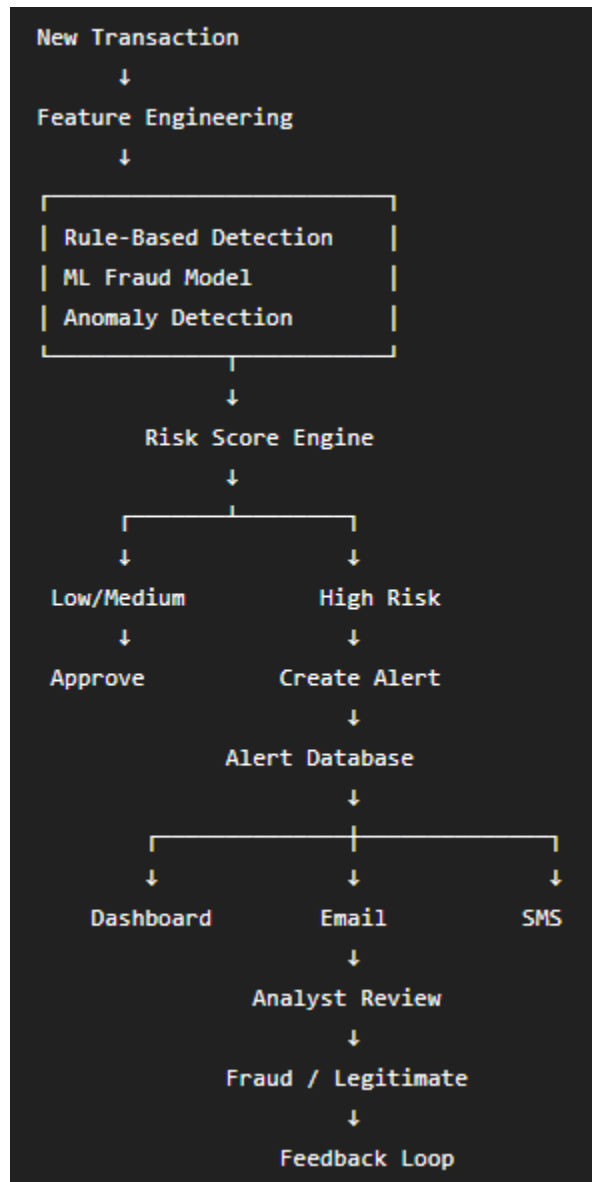
Avoid Duplicate Alerts :

If a customer makes 10 or more transactions at a time, don't need to push notifications for each transactions

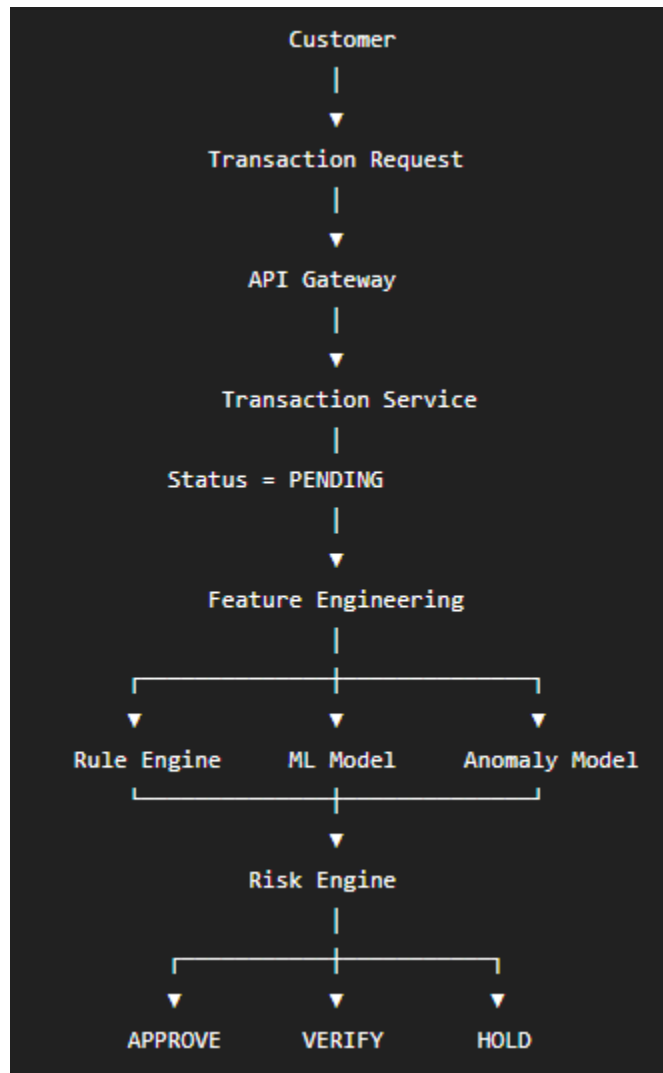
We'll group them like:

{Customer: CUST1001};{ Suspicious Transactions: 10 };{Total Amount: 85,000 BDT }

Final Architecture:{Alert System}



##Analyzed Real-Time Architecture for Fraud Detection



Colab Link:

[Fraud Detection in Bank Transactions](#)

[New dataset Of a bank in India](#)

Datasets: